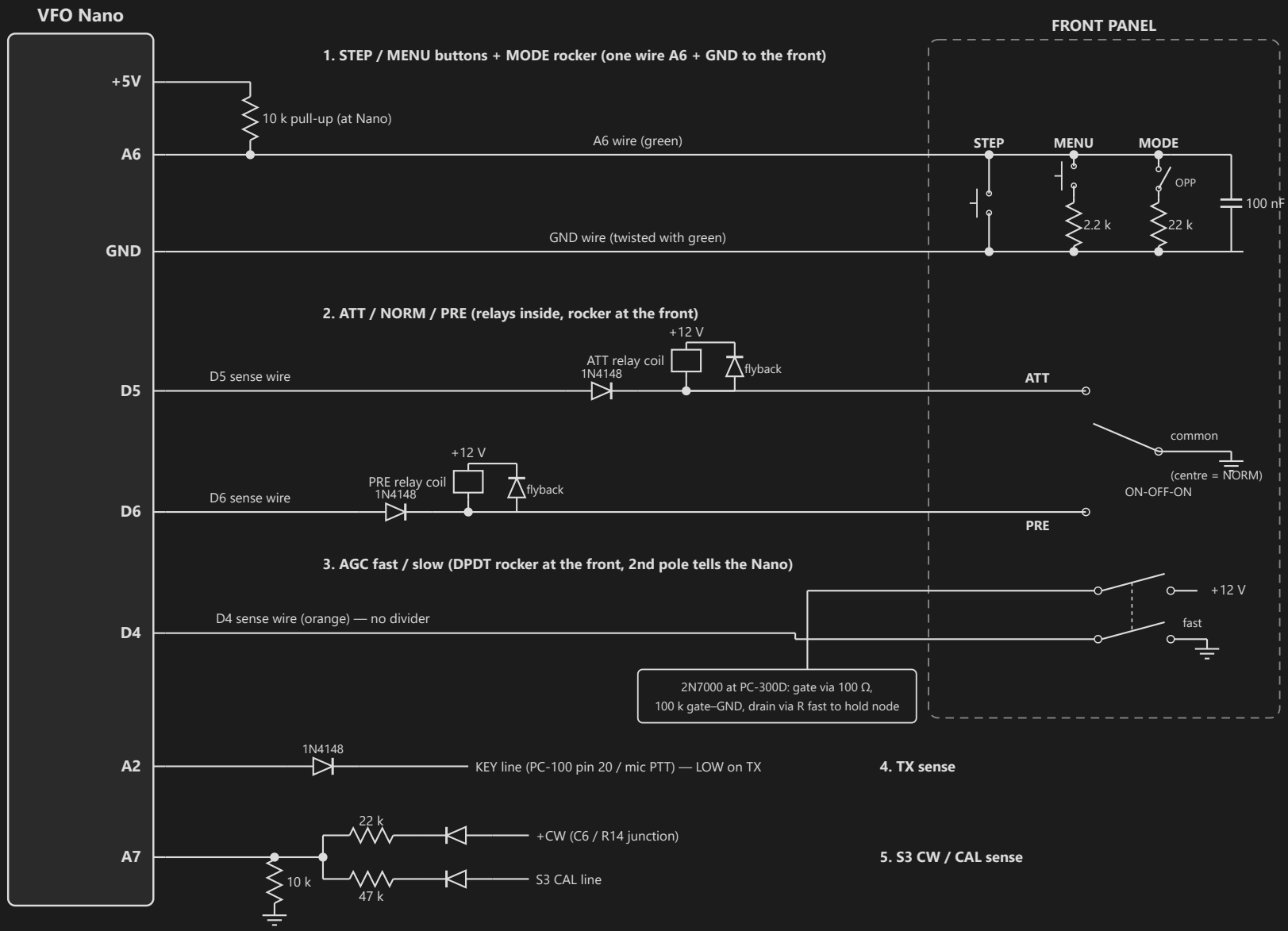
# Atlas 210X — VFO Nano buttons, rockers, relays and sense wires

YT3CT · 2026-10 · proposal · diode band = cathode. Parts at the source end, plain wires to the Nano (5 V or less).




- From the Nano to the front / relays: **A6, GND, D4, D5, D6**. A6 readings: STEP 0, MENU ≈ 184 (170 with MODE), MODE alone ≈ 700, nothing 1023. A2 goes to the KEY line, A7 to the S3 network. The +12 V comes from the radio, never from the Nano.
- D5 / D6 / A2: Nano internal pull-up, LOW = on. D6 becomes PRE sense only after STEP has moved to A6. Relay contacts (RF in / out of the π-pad and preamp) are not shown.
- A7: CW ≈ 3.8 V, CAL ≈ 2.1 V, REC / TRANS 0 V. Set CW_SENSE_FITTED 1 in the sketch once this network is wired.

## Wires leaving the VFO box

Everything to run while the VFO box is open. Parts (diodes, resistors) at the far end unless noted.

| # | Nano pin | Wire | Far end | Parts / notes |
|---|---|---|---|---|
| **Rear socket J4 — twisted 4-wire set** | | | | |
| 1 | RST | grey | J4 pin 1 | 1 µF (105) in series, **at the Nano end** (10 nF is too small against the Nano's own 100 nF) |
| 2 | D1 (TX) | green | J4 pin 2 | to adapter RX |
| 3 | D0 (RX) | white | J4 pin 3 | from adapter TX |
| 4 | GND | black | J4 pin 4 | chassis / Nano GND |
| **Front panel buttons — twisted pair** | | | | |
| 5 | A6 | green | STEP / MENU buttons, MODE rocker | 10 k pull-up A6–5 V at the Nano; STEP 0 Ω, MENU 2.2 k, MODE (OPP) 22 k + 100 nF at the front |
| 6 | GND | twisted with green | buttons / MODE common | also the ATT/PRE rocker common |
| **Sense inputs (Nano only listens, 5 V or less on the wire)** | | | | |
| 7 | D5 | grey | ATT relay coil low end | 1N4148, band to the coil |
| 8 | D6 | red | PRE relay coil low end | 1N4148, band to the coil (after STEP moved to A6) |
| 9 | D4 | orange | AGC rocker 2nd pole | DPDT: 2nd pole common = GND, "fast" throw = D4. No divider |
| 10 | A2 | white | KEY line | 1N4148 fitted **at the Nano end** (band to the KEY line), well insulated — the white wire carries +13 V |
| 11 | A7 | blue | S3 network | network → **orange** to CAL (1N4148 + 47 k), **white/black** to +CW (1N4148 + 22 k), 10 k to GND |
| **Output / supply** | | | | |
| 12 | D13 | white/red | VariBeam Nano D8 (through 10 k) | PTT request from CAT. VariBeam keys only after 300 ms HIGH (bootloader blinks ignored), drives D9 → BC547 → PTT relay (wiring sheet section 10) |
| **RF** | | | | |
| 13 | Si5351 CLK2 | thin coax | old PC-820 cal injection point | leave the receiver end open for now; later 1–2 pF or 47 k in series |

- Not from the VFO box: J4 pins 5–8 (audio via transformers, PTT to the KEY line) and the rocker / relay +12 V wiring.
- Route along the chassis, cross coax at 90°. If a reading jumps on TX: 100 nF from that pin to GND at the Nano.